



Principal Component Analysis

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PCA (Algorithm)

- PCA can be thought of as an unsupervised learning problem.
- The whole process of obtaining principle components from a raw dataset can be simplified in six steps:
 1. Take the dataset consisting of $d+1$ dimensions and ignore the labels such that our new dataset shall become d dimensional.
 2. Compute the *mean* for every dimension of the dataset.
 3. Compute the *covariance matrix* of the dataset.

PCA (Algorithm)



4. Compute *eigenvectors* and the corresponding *eigenvalues*.
5. Sort the eigenvectors by decreasing eigenvalues and choose k eigenvectors with the largest eigenvalues to form a $d \times k$ dimensional matrix W .
6. Transform the samples onto the new subspace.

Example



Example:

Suppose we have 6 points/instances in two dimensions or attributes.

$(X_1, X_2) = (2,1), (3,5), (4,3), (5,6), (6,7), (7,8)$

Example

Example

The formula for computation of final covariance matrix is

$$\text{CoV} = \frac{1}{n-1} \sum_{i=1}^n (X - \mu) (X - \mu)^T$$

Where $X = \begin{bmatrix} X1 \\ X2 \end{bmatrix}$

Example

Example: The steps are

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 6 \\ 7 \end{pmatrix}, \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$\mu_{x_1} = \frac{2+3+4+5+6+7}{6} = 4.5$$

$$\mu_{x_2} = \frac{1+5+3+6+7+8}{6} = 5$$

Example

Example: The steps are

$$\begin{bmatrix} X_1 - \mu_{X_1} \\ X_2 - \mu_{X_2} \end{bmatrix} = \begin{bmatrix} 2 - 4.5 \\ 1 - 5 \end{bmatrix}, \begin{bmatrix} 3 - 4.5 \\ 5 - 5 \end{bmatrix}, \begin{bmatrix} 4 - 4.5 \\ 3 - 5 \end{bmatrix}, \begin{bmatrix} 5 - 4.5 \\ 6 - 5 \end{bmatrix}, \begin{bmatrix} 6 - 4.5 \\ 7 - 5 \end{bmatrix},$$
$$\begin{bmatrix} 7 - 4.5 \\ 8 - 5 \end{bmatrix}$$

$$[X - \mu] = \begin{bmatrix} -2.5 \\ -4 \end{bmatrix}, \begin{bmatrix} -1.5 \\ 0 \end{bmatrix}, \begin{bmatrix} -0.5 \\ -2 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \begin{bmatrix} 1.5 \\ 2 \end{bmatrix}, \begin{bmatrix} 2.5 \\ 3 \end{bmatrix}.$$

Example

Example: The steps are

$$(X - \mu) \cdot (X - \mu)^T$$
$$\begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -2.5 & -4 \end{bmatrix} = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix}$$
$$\begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -1.5 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} -0.5 & -2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 \\ 1 & 4 \end{bmatrix}$$
$$\begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

Example

Example: The steps are

$$\begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 1.5 & 2 \end{bmatrix} = \begin{bmatrix} 2.25 & 3 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2.5 \\ 3 \end{bmatrix} \begin{bmatrix} 2.5 & 3 \end{bmatrix} = \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

Now sum all matrices

$$\Sigma = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix} + \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix} + \dots + \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 17.5 & 22 \\ 22 & 34 \end{bmatrix}$$

Example

Example: The steps are

$$\text{Covariance matrix} = \frac{1}{5} \begin{bmatrix} 17.5 & 22 \\ 22 & 34 \end{bmatrix}.$$

$$A = \begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix}.$$

To find the eigenvalues, substituting the values in

$$AX = \lambda X$$

$$\begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix} X = \lambda X$$

$$\begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix} X - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} X = 0$$

$$\begin{bmatrix} 3.5 - \lambda & 4.4 \\ 4.4 & 6.8 - \lambda \end{bmatrix} X = 0$$

Example

Example: Solving Characteristics Equation

take determinant of the equation.

$$\begin{vmatrix} 3.5-\lambda & 4.4 \\ 4.4 & 6.8-\lambda \end{vmatrix} = 0$$

$$(3.5-\lambda)(6.8-\lambda) - (4.4)^2 = 0$$

$$23.8 - 3.5\lambda - 6.8\lambda + \lambda^2 - 19.36 = 0$$

$$\lambda^2 - 10.3\lambda + 4.44 = 0$$

$$a=1, \quad b=-10.3, \quad c=4.44.$$

putting in equation

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

Example: Solving Characteristics Equation

$$\lambda = \frac{-(-10.3) \pm \sqrt{(-10.3)^2 - 4(1)(4.44)}}{2(1)}$$

$$\lambda = \frac{10.3 \pm 9.398}{2}$$

$$\lambda_1 = 9.849$$

$$\lambda_2 = 0.451$$

$$\begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 9.849 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$3.5x_1 + 4.4x_2 = 9.849x_1 \quad \text{--- (1)}$$

$$4.4x_1 + 6.8x_2 = 9.849x_2 \quad \text{--- (2)}$$

Example



Example

As λ_1 value is greater than λ_2 value and we are selecting only 1 principle component

Then putting λ_1 value into equation $AX = \lambda X$

Example

From equation ①

$$9.849x_1 - 3.5x_1 = 4.4x_2$$

$$6.349x_1 = 4.4x_2$$

$$x_1 = 0.6930x_2$$

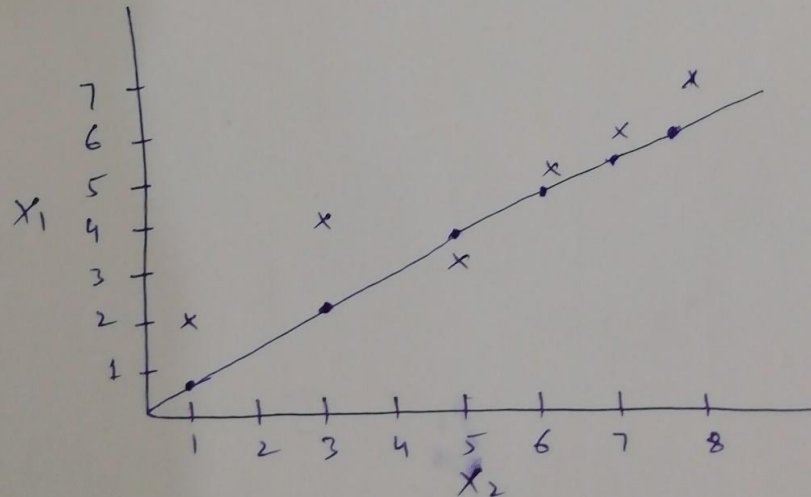
If $x_2 = 1$, then
eigen vector is

$$[0.6930, 1]$$

Example

Using principle component 1.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.6930 \\ 1 \end{bmatrix}, \begin{bmatrix} 3.465 \\ 5 \end{bmatrix}, \begin{bmatrix} 2.079 \\ 3 \end{bmatrix}, \begin{bmatrix} 4.158 \\ 6 \end{bmatrix},$$
$$\begin{bmatrix} 4.851 \\ 7 \end{bmatrix}, \begin{bmatrix} 5.544 \\ 8 \end{bmatrix}$$



Example

Step 4: Form a $d \times k$ dimensional matrix W

$$W = \begin{bmatrix} 0.6930 \\ 1 \end{bmatrix}$$

Step 5:

Now your original features are reduced to

$$X1 = (0.6930, 3.465, 2.079, 4.158, 4.851, 5.54)$$